AI-Driven Smart Fashion Retail Recommendation System

***Abstract*—Fashion retailers operating across multiple stores face challenges in inventory management, promotion targeting, fitting room efficiency, and in store customer assistance. This study presents an integrated artificial intelligence driven Smart Fashion Retail System that combines four intelligent components: inventory and stock flow optimization, personalized marketing intelligence, virtual try on and size recommendation, and location-based customer behavior tracking with staff assistance. The inventory component applies XGBoost regression for demand forecasting and stock transfer recommendations, achieving a coefficient of determination of 0.9497. The marketing component uses XGBoost classification with Recency, Frequency, and Monetary behavioral features to predict promotion redemption, achieving an accuracy of 78.3 percent and a receiver operating characteristic area under the curve of 0.819. The virtual try on component uses Random Forest models for body measurement estimation and brand specific size recommendation, together with Google Cloud Vertex AI for garment visualization. The customer behavior component applies Random Forest classification to movement features to distinguish browsing from transiting and support automated staff assistance. The four components are integrated through FastAPI services and a shared Firebase Firestore database, providing a unified platform for coordinated and data driven decision making in fashion retail.**

Keywords—GPS behavior tracking, Inventory optimization, Personalized marketing, Retrieval-Augmented Generation, Smart retail, Virtual try-on

I. Introduction

AI and machine learning have transformed apparel retail into a real-time, multichannel, data-driven industry [1], [2]. Gradient-boosted trees and random forests support inventory forecasting [3], computer vision and generative models enable virtual try-on [4], RFM/CLV frameworks support targeted promotions [6], and indoor positioning can reveal customer intent through movement trajectories [7]. However, existing solutions often treat forecasting, promotion targeting, tracking, and virtual fitting as separate systems, with limited integration, real-time behavioral input, and explainability [8], [9], [11]. Inventory systems may rely on manual forecasting and stock transfers, while marketing systems use broad segmentation rather than predictive, stock-aware targeting [6], [11]. Indoor tracking can require costly hardware and manual monitoring [7], while virtual try-on often separates measurement estimation from garment visualization [4].

This research proposes an integrated Smart Retail platform combining four modules through a shared FastAPI architecture and Firebase Firestore database: (1) XGBoost-based demand forecasting and explainable stock-transfer recommendations [3], [11]; (2) RFM and XGBoost-based promotion-redemption prediction and promotional asset generation [6]; (3) Random Forest measurement and size prediction with Vertex AI generative try-on and a RAG-based AI stylist [4]; and (4) Random Forest GPS movement classification using Haversine-based features for staff alerts [7], [10]. The study aims to improve multi-store demand forecasting, stock transfers, personalized promotions, measurement-aware virtual try-on, brand-specific sizing, and timely staff assistance.

II. Literature Review

The apparel industry is a major contributor to Sri Lanka's economy through industrial exports and employment [24],[25]. Retail forecasting supports inventory and operational decisions [22], [23], with XGBoost effectively capturing non-linear sales patterns and sparse features [3]. Virtual fitting technologies such as VITON address fit uncertainty through realistic garment overlays [4], while newer virtual dressing-room systems support intelligent clothing selection and purchasing [27]. Customer Relationship Management (CRM) and RFM analysis support customer segmentation and Customer Lifetime Value (CLV) estimation [5], [6], while Random Forest provides accurate classification for large behavioral datasets [10]. Traditional recommendation systems use association rule mining and market basket analysis to identify co-purchasing patterns [12].

Deep convolutional neural networks have provided a foundation for computer-vision-based fashion applications [13], while feature-preserving alignment improves the quality of synthesized garments by retaining textures, logos, and patterns [11]. To address the black-box nature of deep learning, SHAP and LIME provide interpretable explanations for model decisions [8], [9]. Neural Factorization Machines combine deep learning with feature interactions for recommendation [14], while visually-aware systems use generative models to recommend products based on visual similarity and style [15]. LLMs such as GPT-4 further enable natural-language understanding, product content generation, and conversational shopping [16]. Production deployment requires MLOps platforms such as Google Cloud Vertex AI for model training, deployment, and monitoring [17].

Foundational recommendation research covers collaborative filtering, content-based methods, and hybrid architectures [1], with matrix factorization remaining a key approach for learning latent user-item representations [2]. Multimodal models such as Gemini 1.5 Flash enable fast visual-text reasoning for outfit analysis [18], while Imagen 3 supports high-fidelity apparel image generation [19]. Firebase Firestore provides synchronized cloud storage for user profiles, carts, and recommendation data [20], while uplift modeling helps identify persuadable customer segments for targeted promotions [21].

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III. METHODOLOGY, SYSTEM DESIGN AND IMPLEMENTATION

A. Proposed Smart Retail System

The proposed Smart Retail System is an integrated AI-driven platform designed to enhance retail operations and customer experience through intelligent decision-making. It comprises four interconnected modules: (1) AI-Driven Smart Inventory and Stock-Flow Optimization, (2) AI-Driven Personalized Marketing Intelligence, (3) AI-Based Virtual Try-On and Size Recommendation, and (4) GPS-Based Customer Behavior Tracking with Staff Assistance. Modern AI-driven retail platforms emphasize integrated intelligent services to improve operational efficiency and customer engagement [1], [2]. Accordingly, the proposed system adopts a unified architecture that enables seamless information exchange among all modules, supporting coordinated and data-driven retail decision-making as illustrated in Fig 1.

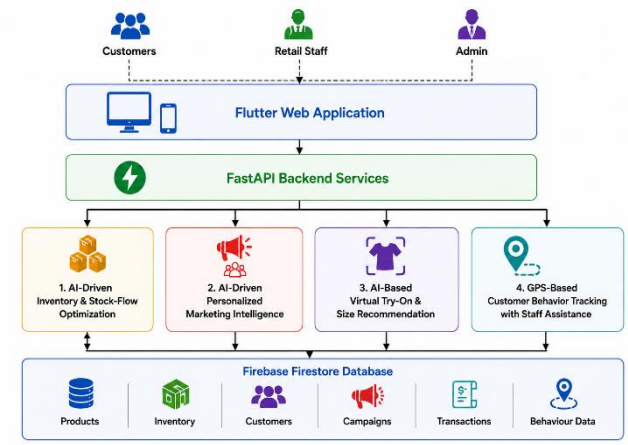


Fig. 1. Overall architecture of the smart retail system.

The system was developed through requirement analysis, data collection, preprocessing, AI model development, system integration, and evaluation as mentioned in Fig 2.

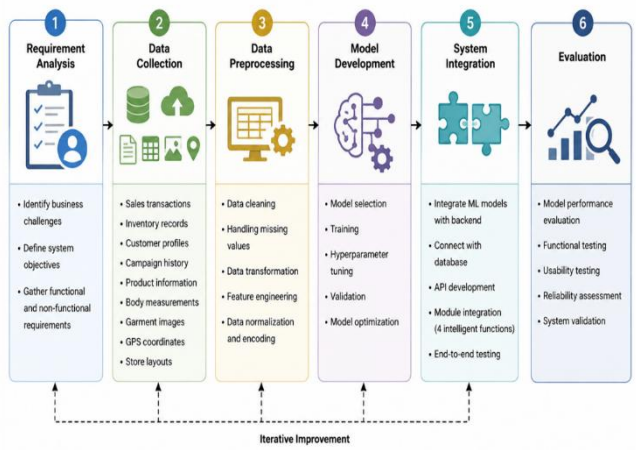


Fig. 2. Research methodology of the smart retail system.

Domain-specific datasets were processed using feature engineering, and the integrated system was validated through functional testing and performance evaluation to ensure reliability and interoperability [3].

B. AI-Driven Smart Inventory and Stock-Flow Optimization

This component predicts future product demand and recommends stock movement across multiple retail stores. Demand is forecast using an XGBoost Regressor trained on approximately 1,000 products across six brands. The model uses historical, lag-1 and lag-7 sales, seven-day rolling averages, product category, subcategory, brand, gender, price, promotion percentage, store, weather conditions, and calendar variables.

The predicted demand is compared with current stock to determine inventory-health status, reorder requirements, and days-on-hand. When a shortage is detected, a smart stock-recommendation engine ranks candidate source stores using available stock, predicted demand, safety-stock requirements, transfer distance, logistics cost, and source-store shortage risk. The resulting recommendation includes confidence and risk scores and follows a review–approve–reject–cancel–execute workflow. After execution, inventory, transaction records, and movement history are updated automatically. A Gemini-based explainability layer converts the technical outputs into natural-language explanations for retail managers.

For implementation, the backend was developed using FastAPI with endpoints for forecasting, decision generation, reference data, product and store information, AI-generated explanations, and recommendation management. The Flutter Web frontend provides forecasting, inventory-health, source-store comparison, cost and distance information, risk and confidence indicators, explanations, and transfer approval controls. Firebase Firestore stores product, store, inventory, sales, forecast, recommendation, transaction, and movement-history data.

C. AI-Driven Personalized Marketing Intelligence

This component supports personalized promotion of products to individual customers in a multi-store fashion retail environment. Customer purchase and campaign-interaction history is used to calculate Recency, Frequency, and Monetary (RFM) features [6]. These features, together with product, promotion, and customer attributes, are provided to an XGBoost classification model [3] to predict the probability that a customer will redeem a given promotional offer.

The predicted redemption probability supports customer segmentation, promotion scheduling, and personalized campaign decisions. A generative poster module uses Google's Gemini Imagen 3 API [19] to create promotional visual assets for high-affinity customer segments, while Gemini 1.5 Flash [18] generates natural-language explanations for recommendations. Product availability and

overstock information are shared with the inventory component to ensure that promoted products are available.

The marketing component was implemented as an analytics dashboard backed by the XGBoost prediction pipeline. The implementation includes CSV upload and data-cleaning, customer-intelligence analysis, promotion scheduling, promotional-poster generation, Firebase Firestore integration [20], and an explainability interface. The implementation therefore connects predictive marketing decisions with the shared retail data infrastructure.

D. AI-Based Virtual Try-On and Size Recommendation

This component functions as a digital fitting room that combines body-measurement prediction, brand-specific size recommendation, and virtual garment visualization. A Random Forest Regressor, optimized using GridSearchCV with three-fold cross-validation, predicts shoulder width, waist circumference, and leg length from height, weight, and gender, with BMI used as an additional feature. A 100-tree Random Forest Classifier maps the predicted standard size to a brand-specific size using categorical product attributes.

When a customer uploads a photograph, the photograph and selected garment image are sent to Google Cloud Vertex AI's virtual try-on model to generate the fitted garment visualization. The resulting image and recommended brand size are returned to the customer-facing application. Changes to stored height or weight automatically trigger recalculation of predicted body measurements.

The implementation uses separate services for body-measurement prediction, brand-size classification, and generative image processing. The Flutter frontend captures customer measurements, performs unit conversion, retrieves predicted sizes, and manages the customer and garment images used for virtual try-on. Firebase Firestore stores customer profile information and predicted measurements.

The system also includes a conversational AI stylist. Product metadata, descriptions, and customer reviews are embedded using the HuggingFace all-MiniLM-L6-v2 model and stored in ChromaDB. LangChain retrieves the most relevant in-stock products using vector similarity, and the retrieved information is provided to the language model to generate product-grounded styling recommendations.

E. GPS-Based Customer Behavior Tracking and Staff Assistance

This component monitors customer movement inside a store using GPS information from mobile devices. GPS coordinates are transmitted at three-second intervals and mapped to predefined three-dimensional store zones. The system calculates movement distance using the Haversine formula together with velocity and direction-change features. These features are provided to a 100-tree Random Forest Classifier to classify customers as "Browsing" or "Transiting." A rule-based speed fallback is used when the trained model is unavailable.

An automated alerting engine monitors dwell time and zone-transition history. Continuous browsing for 30 seconds or more, or a pacing pattern such as Zone A → Zone B → Zone A within 60 seconds, generates a staff-assistance alert. Alerts and customer locations are displayed on a staff monitoring dashboard.

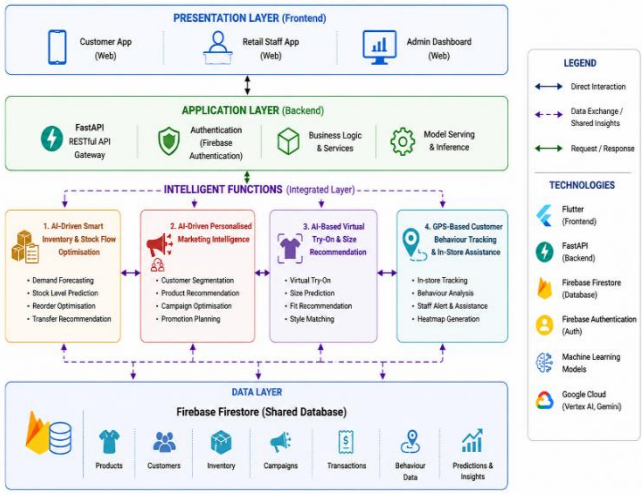


Fig. 3. System integration of the smart retail system.

The implementation consists of a FastAPI monitoring router, zone-spatial service, trajectory machine-learning service, and alerting engine. The Flutter customer application transmits GPS coordinates every three seconds and provides a manual SOS option. A separate staff dashboard polls the backend to display customer locations and pending assistance alerts. Firebase Firestore stores zone information, customer monitoring data, assistance requests, and staff notifications.

F. Technologies Used

The proposed Smart Retail System integrates artificial intelligence and machine learning to support four intelligent modules within a unified retail platform. The technologies used for the relevant system layer is as follows,

TABLE I. TECHNOLOGIES USED IN THE SMART RETAIL SYSTEM

System Layer	**Technology**
Frontend	Flutter Web
Backend	FastAPI (Python)
Database	Firebase Firestore
Authentication	Firebase Authentication
Machine Learning	XGBoost, Random Forest (Scikit-learn)
AI Services	Gemini (explanations and image generation), Google Cloud Vertex AI (virtual-try-on-001), LangChain with ChromaDB and HuggingFace all-MiniLM-L6-v2 (conversational AI stylist RAG pipeline)
Programming Languages	Python, Dart
Other Tools	Pandas, NumPy, GridSearchCV, GitHub Pages (image hosting)

G. System Integration

The four modules communicate through shared backend services and a centralized Firebase Firestore database. Common identifiers, including identifiers for each product,

customer and transaction support information exchange across the system as illustrated in Fig 3.

IV. RESULTS AND DISCUSSION

The evaluation of the integrated AI-based Smart Retail System demonstrates strong predictive performance across the four core operational components.

A. Inventory Forecasting and Stock-Flow Optimization

The daily XGBoost regression model recorded a Mean Absolute Error (MAE) of 0.5418, a Root Mean Squared Error (RMSE) of 0.9436, and a coefficient of determination (R^2) of 0.9497 across eight store branches. The monthly inventory-intelligence model, trained on approximately 297,011 records spanning 1,000 products and eight stores, achieved an MAE of 14.213, an RMSE of 26.306, and an R^2 of 0.9353. Feature-importance analysis showed that recent and historical demand variables were the dominant predictors, while promotional, seasonal, customer, and environmental factors provided additional predictive information. The forecasting outputs were subsequently used for shortage detection and inter-store stock- transfer recommendations as reflected in Fig4.

TABLE II. INVENTORY FORECASTING MODEL PERFORMANCE

Model	MAE	RMSE	R^2
Daily demand forecasting (XGBoost)	0.5418	0.9436	0.9497
Monthly inventory intelligence (XGBoost)	14.213	26.306	0.9353

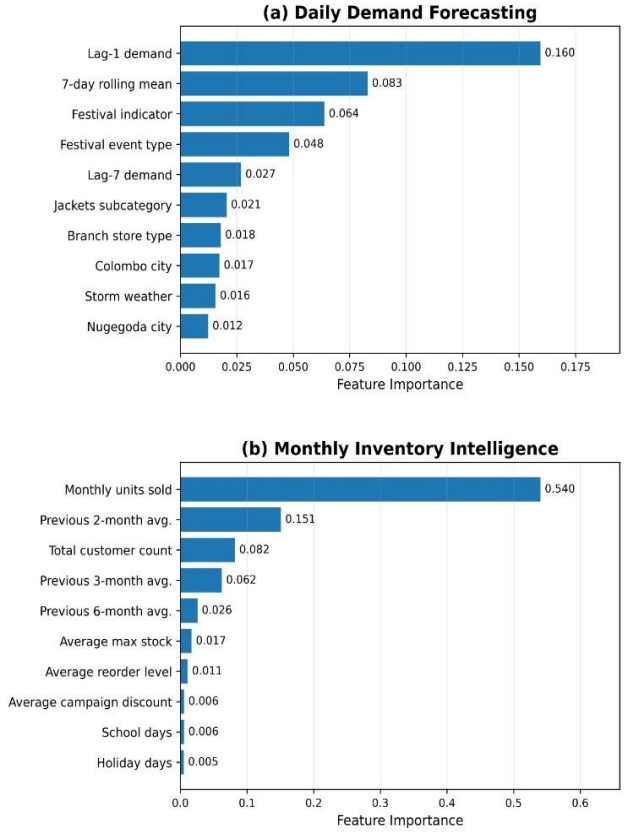


Fig. 4. Feature importance of the inventory forecasting models: (a) daily demand forecasting and (b) monthly inventory intelligence

B. Personalized Marketing Intelligence

The XGBoost redemption classifier achieved a test accuracy of 85.5% and a ROC-AUC of 0.819, substantially outperforming the 35.9% baseline redemption rate. This predictive uplift means marketing teams can accurately target high-affinity customer segments, reduce wasted promotional expenditure while optimizing campaign timing around seasonal salary cycles. Switching the prediction target from offer-type classification to redemption-probability prediction was found to be the single largest contributor to this accuracy improvement. Fig. 5 and Fig. 6 present the confusion matrix and ROC curve underlying these results.

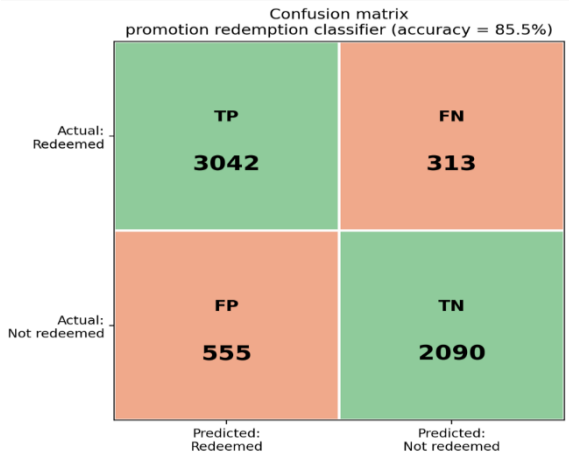


Fig. 5. Confusion matrix for the promotion redemption classifier

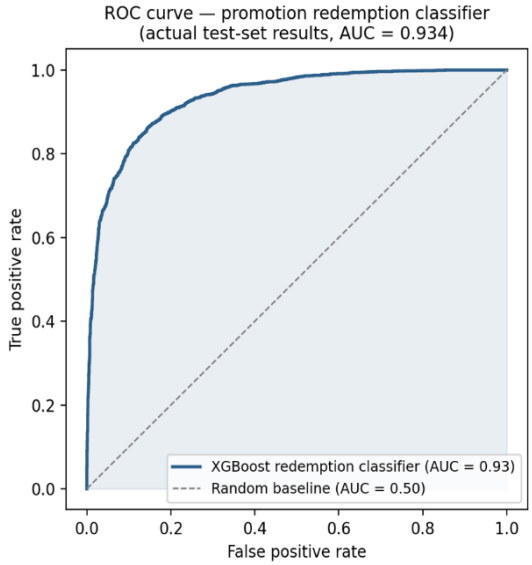


Fig. 6. ROC curve for the promotion redemption classifier

C. Virtual Try-On and Size Recommendation

The Virtual Try-On and Size Recommendation component uses two machine learning models to support personalized sizing. The body measurement prediction model, based on multi-output regression, predicts shoulder width, waist circumference, and leg length using height, weight, gender, and BMI. As reflected in Fig 7 model achieved MAE values of 1.12 cm, 1.45 cm, and 1.28 cm,

with R^2 scores of 0.914, 0.897, and 0.921, respectively. The brand-specific sizing model classifies a customer's standard size into a corresponding brand-specific size using historical fit data. It achieved an overall classification accuracy of 88.4%, with weighted precision, recall, and F1-score of 0.879, 0.884, and 0.881, respectively.

TABLE III. VIRTUAL TRY-ON AND SIZE RECOMMENDATION MODEL PERFORMANCE (BRAND SPECIFIC SIZING MODEL)

Metric	Value
Overall Classification Accuracy	88.4%
Precision (Weighted)	0.879
Recall (Weighted)	0.884
F1-Score (Weighted)	0.881

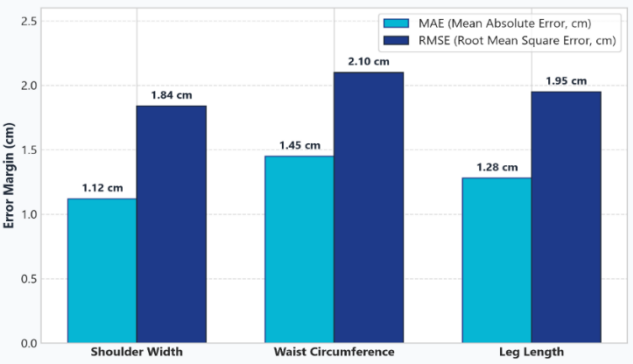


Fig. 7. Body Measurement Regression Error Breakdown.

D. Customer Behavior Tracking

The Customer Trajectory and Intent Tracing Model, integrated within the Customer Behavior Tracking and Real Time Staff Assistance system, utilizes Random Forest classification to predict the customer's likelihood to purchase or their specific shopping intent based on their real-time movement, zone dwell times, and physical interactions. The classification model achieved an overall accuracy of 91.2%. Comprehensive performance evaluation demonstrated a weighted precision of 0.908, a weighted recall of 0.912, and a weighted F1-score of 0.910. Furthermore, as illustrated in Fig 8 , the model recorded an ROC-AUC score of 0.895, indicating strong discriminative capability.

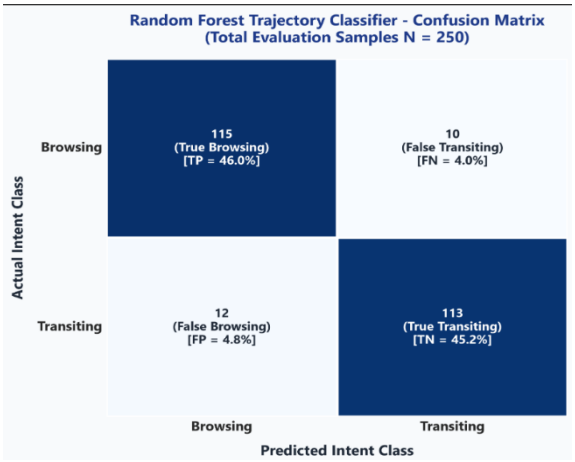


Fig. 8. Confusion matrix for the Random Forest Trajectory classifier

V. RESEARCH CONTRIBUTION AND LIMITATIONS

The overall system inherits the limitations associated with its individual components, including data dependency, reliance on partially simulated datasets, and operational constraints.

A. Inventory Optimization

The inventory component contributes an integrated inventory-decision workflow that extends beyond demand forecasting by linking predictions to explainable, risk-aware stock-transfer recommendations and execution support. This enables more informed inventory management through source-store selection, risk assessment, and logistics-cost estimation. However, the model relies primarily on simulated or prepared retail data, simplified logistics assumptions, and still requires human approval before transfer execution.

B. Personalized Marketing Intelligence

The marketing component contributes a closed-loop marketing-intelligence framework that integrates redemption-probability prediction, generative promotional-asset creation, and stock availability information to support more effective customer targeting and campaign delivery. However, the model's performance depends on the quality and completeness of campaign and customer data, while the generated promotional assets and explanations are constrained by the information provided to the underlying generative model.

C. Virtual Try-On

The virtual try-on component contributes an integrated pipeline that combines anthropometric regression, brand-sizing classification, generative image compositing, and a RAG-grounded conversational AI stylist to support automated size recommendation, virtual garment visualization, and inventory-grounded styling advice. This streamlines the apparel-selection process by reducing reliance on manual measurements and brand-specific size charts, while the retrieval-augmented stylist reduces the risk of hallucinated or generic recommendations from the underlying LLM. However, the system may produce visual artefacts for complex poses or obscured images, depends on accurate user-provided height and weight information, is less accurate for brands that are not represented in the training data, and the stylist's advice quality depends on the completeness of the product metadata indexed in the vector database.

D. GPS-Based Tracking

The GPS tracking component contributes an automated behavioral-alerting framework that combines standard mobile GPS hardware, geometric tolerance handling, and a lightweight Random Forest classifier to identify customer behavioral patterns without requiring dedicated beacon or camera infrastructure. This enables location-aware behavioral monitoring using widely available mobile devices. However, GPS accuracy may degrade in indoor environments, continuous background location tracking can significantly impact battery life, and the classifier may require retraining with real-world movement data before deployment. Additionally, the system depends on users keeping the store application active in the background.

VI. CONCLUSION AND FUTURE WORK

This research developed an integrated optimization system for multi-store apparel retail, combining mobile, cloud, machine learning, generative AI, and communication technologies. The platform unifies demand forecasting and inventory optimization, personalized marketing, virtual try-on with size recommendation, and real-time customer behavior tracking, using edge and cloud computing for efficient processing and timely decisions.

Future work includes improving forecasting with hybrid ML models and additional data sources (e.g., live traffic, POS data); expanding virtual try-on with larger training datasets and fine-tuned generative models; exploring alternative indoor-positioning technologies alongside GPS; and enabling adaptive, continuously learning models across all four components. Full-platform deployment and real-world testing will further validate the system's effectiveness and scalability.

REFERENCES

[1] F. Ricci, L. Rokach, and B. Shapira, Recommender Systems Handbook, 3rd ed. Springer, 2022.
[2] Y. Koren, R. Bell, and C. Volinsky, "Matrix Factorization Techniques for Recommender Systems," Computer, vol. 42, no. 8, pp. 30–37, Aug. 2009.
[3] T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," in Proc. ACM SIGKDD, 2016, pp. 785–794.
[4] X. Han, Z. Wu, Z. Wu, R. Yu, and L. S. Davis, "VITON: Image-Based Virtual Try-On Network," in Proc. IEEE CVPR, 2018.
[5] V. Kumar and W. Reinartz, Customer Relationship Management, 3rd ed. Springer, 2018.
[6] P. S. Fader, B. G. S. Hardie, and K. L. Lee, "RFM and CLV: Using Customer Lifetime Value for Better Marketing Decisions," Journal of Marketing Analytics, vol. 3, no. 4, pp. 220–230, 2015.
[7] J. Krumm, "A Survey of Computational Location Privacy," Personal and Ubiquitous Computing, vol. 13, no. 6, pp. 391–399, 2009.
[8] S. M. Lundberg and S.-I. Lee, "A Unified Approach to Interpreting Model Predictions," in Advances in Neural Information Processing Systems (NeurIPS), 2017.
[9] M. T. Ribeiro, S. Singh, and C. Guestrin, "Why Should I Trust You? Explaining the Predictions of Any Classifier," in Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining (KDD), 2016, pp. 1135–1144.
[10] L. Breiman, "Random Forests," Machine Learning, vol. 45, no. 1, pp. 5–32, 2001.
[11] B. Wang et al., "Characteristic-Preserving Image-Based Virtual Try-On Network," in Proc. ECCV, 2018.
[12] M. H. Dunham, Data Mining: Introductory and Advanced Topics. Pearson, 2003.
[13] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet Classification with Deep Convolutional Neural Networks," Communications of the ACM, vol. 60, no. 6, pp. 84–90, 2017.
[14] X. He and T.-S. Chua, "Neural Factorization Machines," in Proc. ACM SIGIR, 2017.
[15] Y. Kang et al., "Visually-Aware Fashion Recommendation with Generative Image Models," IEEE Transactions on Multimedia, vol. 23, pp. 3445–3458, 2021.
[16] OpenAI, "GPT-4 Technical Report," 2023.
[17] Google Cloud, "Vertex AI Documentation," Google Cloud, 2024.
[18] Google, "Gemini 1.5 Flash," Google AI for Developers, 2024.
[19] Google, "Imagen 3," Google AI for Developers, 2024.
[20] Google, "Firebase Firestore Documentation," 2024.
[21] Y. T. Park, T. Xu, and M. Anany, "Enhancing uplift modeling in multi-treatment marketing campaigns: Leveraging score ranking and calibration techniques," arXiv preprint arXiv:2408.13628, 2024.
[22] S. Makridakis, E. Spiliotis, and V. Assimakopoulos, "The M4 Competition: Results, Findings, Conclusion and Way Forward," International Journal of Forecasting, vol. 34, no. 4, pp. 802–808, 2018.
[23] R. Fildes, M. Ma, and S. Kolassa, "Retail Forecasting: Research and Practice," International Journal of Forecasting, vol. 35, no. 1, pp. 1–9, 2019.
[24] Sri Lanka Export Development Board, Sri Lanka Apparel Industry Report, 2024
[25] Central Bank of Sri Lanka, Annual Economic Review, 2025.
[26] S. W. P. N. M. Weerasinghe, R. M. D. D. Rajapaksha, L. G. I. Sathsara, H. S. D. N. Gunasekara, D. R. Wijendra, and D. I. De Silva, "Virtual Dressing Room: Smart Approach to Select and Buy Clothes," in *2021 3rd International Conference on Advancements in Computing (ICAC)*, 2021, pp. 193–198, doi: 10.1109/ICAC54203.2021.9671198.